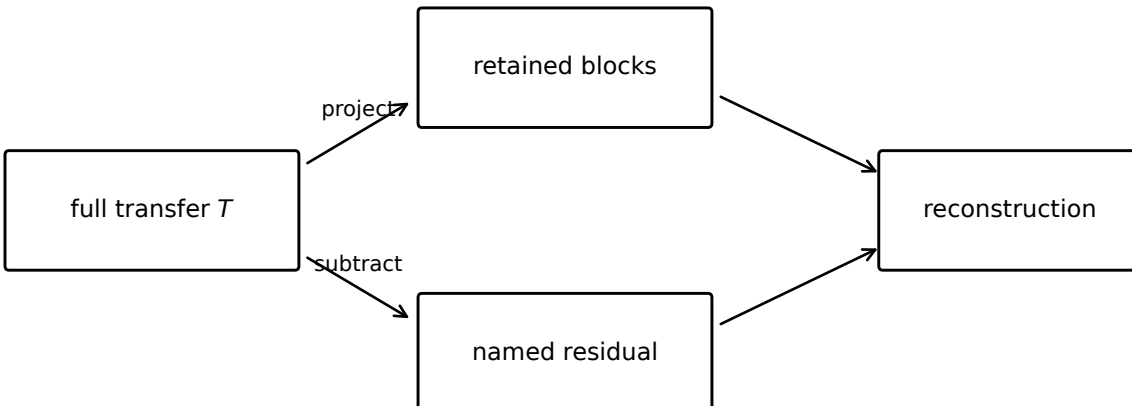compression is safe only when the complement remains explicit